

EEP 596 Computer Vision - Assignment 5 Report

PyTorch and Backpropagation

Task 1: Chain Rule

For function $f(x,y,z) = xy+z$, compute df/dz , df/dq , df/dx , and df/dy where $q=xy$.

Input values: $x=-2$, $y=5$, $z=-4$

Derivative	Value
df/dz	1.0
df/dq	1.0
df/dx	5.0
df/dy	-2.0

Task 2: ReLU Backpropagation

For input data= $[-1,-2]$ and weights= $[2,-3,-3]$, compute dx and dw after backpropagation.

Gradient w.r.t. inputs (dx): $[2, -3]$
Gradient w.r.t. weights (dw): $[-1, -2, 1]$

Task 3: Chain Rule with PyTorch

Part (a): Forward Pass Values

Calculate a, b, c values for $f(w,x) = 1/(1 + \exp(-(w_0x_0+w_1x_1+w_2)))$

Variable	Value
a $(\exp(-s_3))$	0.3679
b $(a + 1)$	1.3679
c $(1/b)$	0.7311

Part (b): Backward Pass Gradients

Gradient	Value
$\partial c / \partial w_0$	-0.19660000503063202
$\partial c / \partial x_0$	0.39320001006126404
$\partial c / \partial w_1$	-0.39320001006126404
$\partial c / \partial x_1$	-0.5898000001907349
$\partial c / \partial w_2$	0.19660000503063202

Task 4: Backpropagation with tanh

For $f(w,x) = \tanh(w_0x_0+w_1x_1+w_2)$ with $w=[5,2]$, $x=[-1,4]$, $w_2=-2$, and MSE loss with ground truth=1

Part (a): Forward Pass

Output $\hat{y} = f(\mathbf{w}, \mathbf{x})$: 0.761594

Part (b): Gradients

Weight	Gradient
$\partial L / \partial w_0$	0.200249
$\partial L / \partial w_1$	-0.800995
$\partial L / \partial w_2$	-0.200249

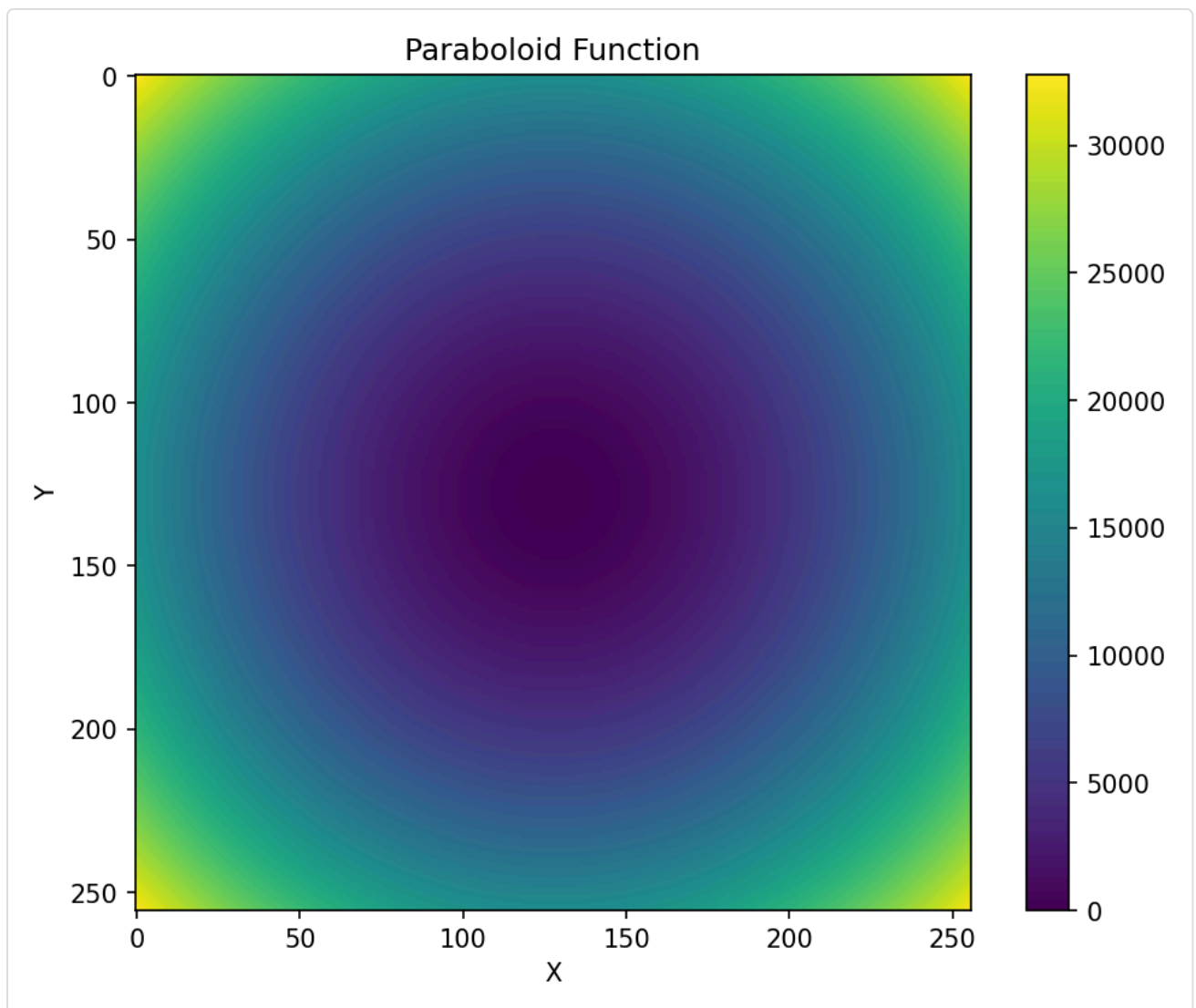
Part (c): Updated Weights (learning rate = 0.1)

Weight	Original	Updated
w_0	5.0	4.979975
w_1	2.0	2.080099
w_2	-2.0	-1.979975

Task 5: Optimization Methods

Finding the minimum of a 2D paraboloid using different optimization methods.

Paraboloid Visualization

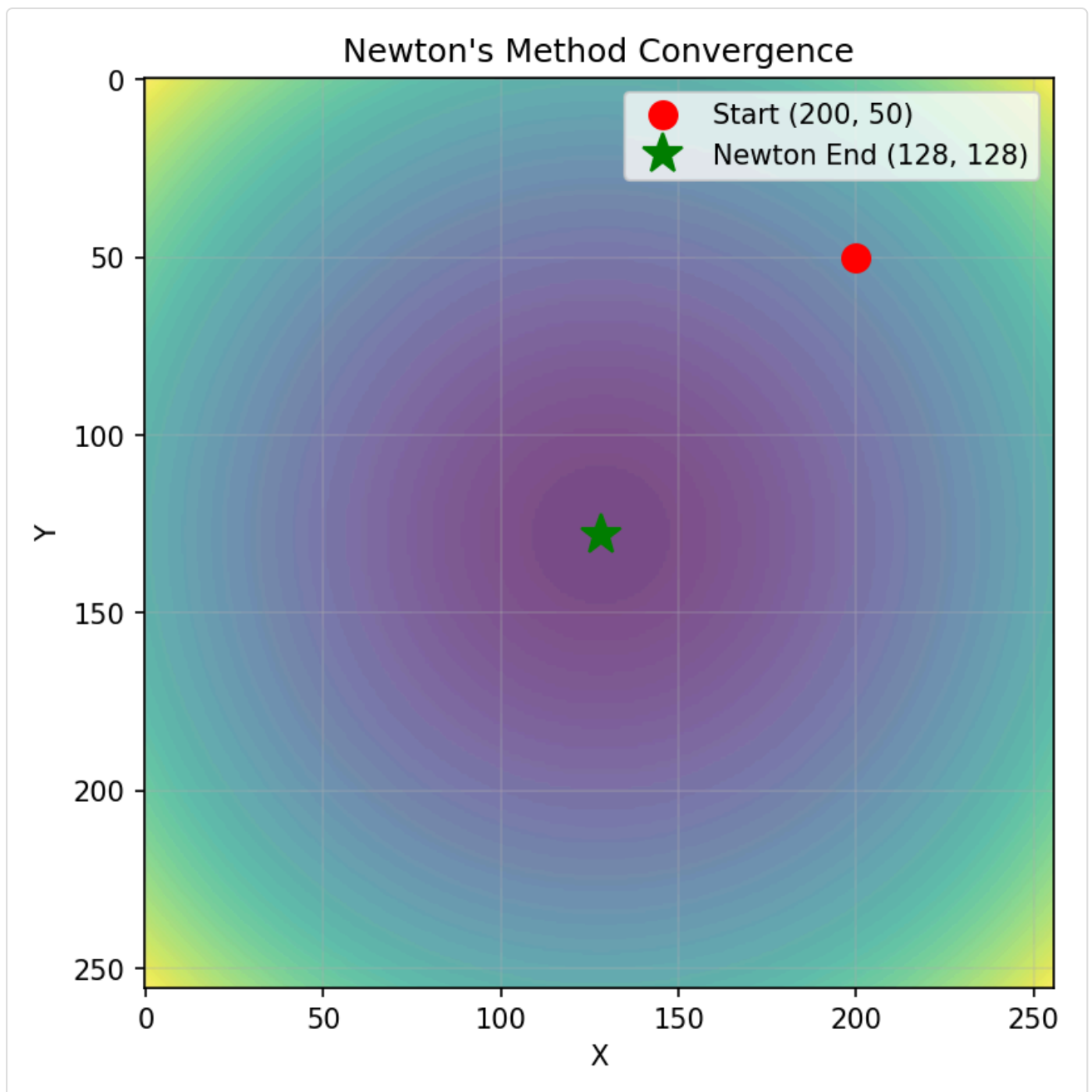


Part (a): Newton's Method

Starting from position (200, 50)

Converged to: (128, 128)

Iterations to converge: 2

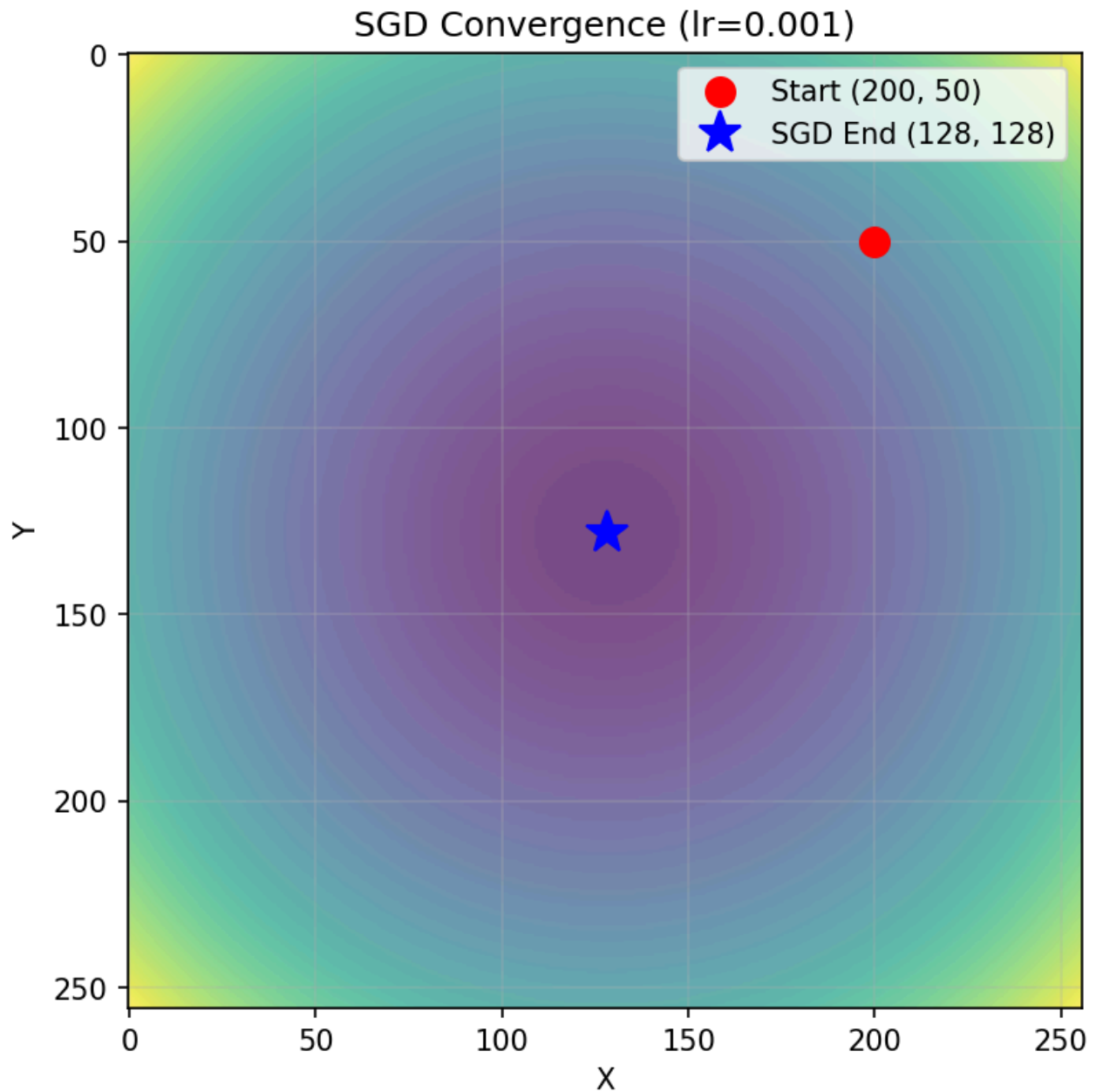


Part (b): Stochastic Gradient Descent (SGD)

Starting from position (200, 50) with learning rate = 0.001

Converged to: (128, 128)

Iterations to converge: 2467



Learning Rate Analysis

1) Effect of learning rate:

- **Small learning rate (e.g., 0.00001):** Slower convergence but more stable
- **Medium learning rate (e.g., 0.001):** Good balance between speed and stability
- **Large learning rate (e.g., 1.0):** May converge faster or cause divergence/oscillation

2) Divergence:

A learning rate that is too large (e.g., > 1.0) can cause divergence, where the optimization overshoots the minimum and the values oscillate or grow without bound. For this

paraboloid, learning rates above approximately 1.0-2.0 may cause divergence depending on the starting position.

Learning Rate Experiments

Learning Rate	Final Position	Iterations to Converge
1e-05	(193, 57)	5000
0.001	(128, 128)	2467
1.0	(200, 50)	5000